



Department of Computer Science and Engineering
Bangladesh University of Engineering and Technology

— CSE 450 CAPSTONE · FINAL PRESENTATION

Antiphony

Dual-perspective modeling of persuasion and
commitment in asymmetric dialogue

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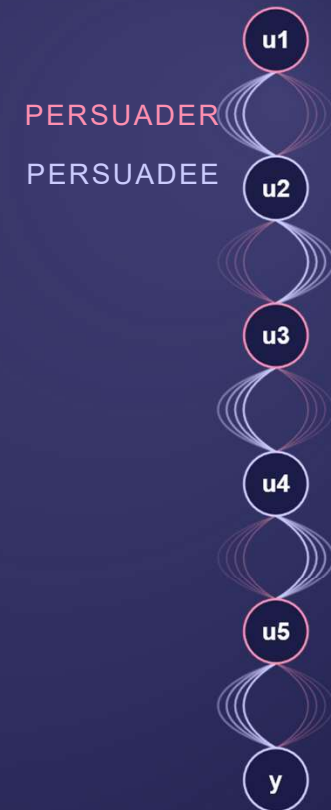
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— PART 1 OF 5

Problem

1.1 The label disagrees with the transcript

1.2 Problem statement

1.3 Why it's hard

1.4 What we built



The Label Disagrees with the Transcript

P4G DIALOGUE #124

PERSUASION FOR GOOD

Persuader

“We have you down for a \$10 donation. Is that correct?”

Persuadee

“Yes that is.”

Recorded donation field

\$0

WHY IT MATTERS

In Bangladeshi social commerce, sales happen inside chats (Messenger, WhatsApp). Pathao needed two things:

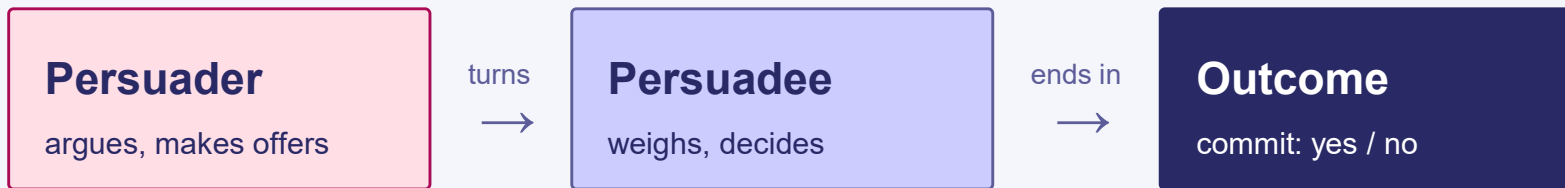
- 01 Triage of high-intent leads in real time
- 02 A small model that runs on their own servers

NOT A ONE-OFF

Dialogue #842: “I will absolutely look to donating something later.” Both donation fields show \$0.

“You cannot learn commitment from a label that’s wrong, or from a model that never reads the end of the chat.”

Problem Statement: Three Sub-Problems



SP1 · INTENTION

Intention

Given dialogue $D = (u_1 \dots u_N)$, predict commitment $y \in \{\text{yes}, \text{no}\}$.

One label per dialogue, from the decider's final position

SP2 · STRATEGY

Strategy

For each persuader turn, predict its strategy set.

11 categories, 41 strategies, multi-label

SP3 · TRANSFER

Transfer

In buyer–merchant negotiation, does the model follow the trajectory or just read the last line?

$\Delta_{\text{suff}} = \text{Macro-F1}(\text{full dialogue}) - \text{Macro-F1}(\text{last turn})$

Why It's Hard

01 · TRUNCATION

0.70

Macro-F1 cap for flat models
256-token models see only about the first half of a dialogue (median 344 words).

02 · SPEAKER BLUR

2 → 1

speakers flattened into one text

The model has to rediscover who the decider is.

03 · LONG TAIL

2

turns of door_in_the_face
conditional has 13 examples; 48.9% of turns are multi-label.

04 · SHORTCUTS

“sounds good”

a single closing line
can give the answer away without reading the dialogue.

COMPLEX ENGINEERING PROBLEM · TABLE 1.1

P1 · Depth of knowledge

Loss design, hierarchical attention, recurrent state, calibrated ensembles

P2 · Conflicting requirements

2,048-token dialogues against a 16 GB T4 memory ceiling

P3 · Depth of analysis

Monolithic transformers failed; a new dual-stream network was needed

P4 · Infrequent issues

2 to 5 instance tail classes; silent truncation of 30.4% of endings

P5 · No standard codes

Custom decision tests and three-phase human verification

P6 · Stakeholders

Merchant conversion against consumer protection and privacy

P7 · Interdependence

Pass 3 labels forced retraining of the whole 14-model lineage

What We Built

Intention

P4G, 1,017 dialogues, re-annotated in 3 passes

0.862

Antiphony Macro-F1 · strict labels
0.845

Strongest of the fourteen model versions we built

Strategy

10,600 turns, 11 categories / 41 strategies

0.706

micro-F1 = 89% of the 0.79 human ceiling

19-member ensemble with a calibrated blend

Transfer

700 synthetic negotiations (500 + 200 reversals)

+0.0384

Δ_{suff} , up from +0.0001

Antiphony retrained on buyer–merchant chats

Next: how the three pieces fit together

— PART 2 OF 5

System and Data

2.1 System overview

2.2 Intention ground truth

2.3 Strategy taxonomy and annotation

2.4 Corpus statistics

2.5 Negotiation benchmark



System Overview

ROWS · THREE STREAMS

3a **Intention** · P4G donation commitment

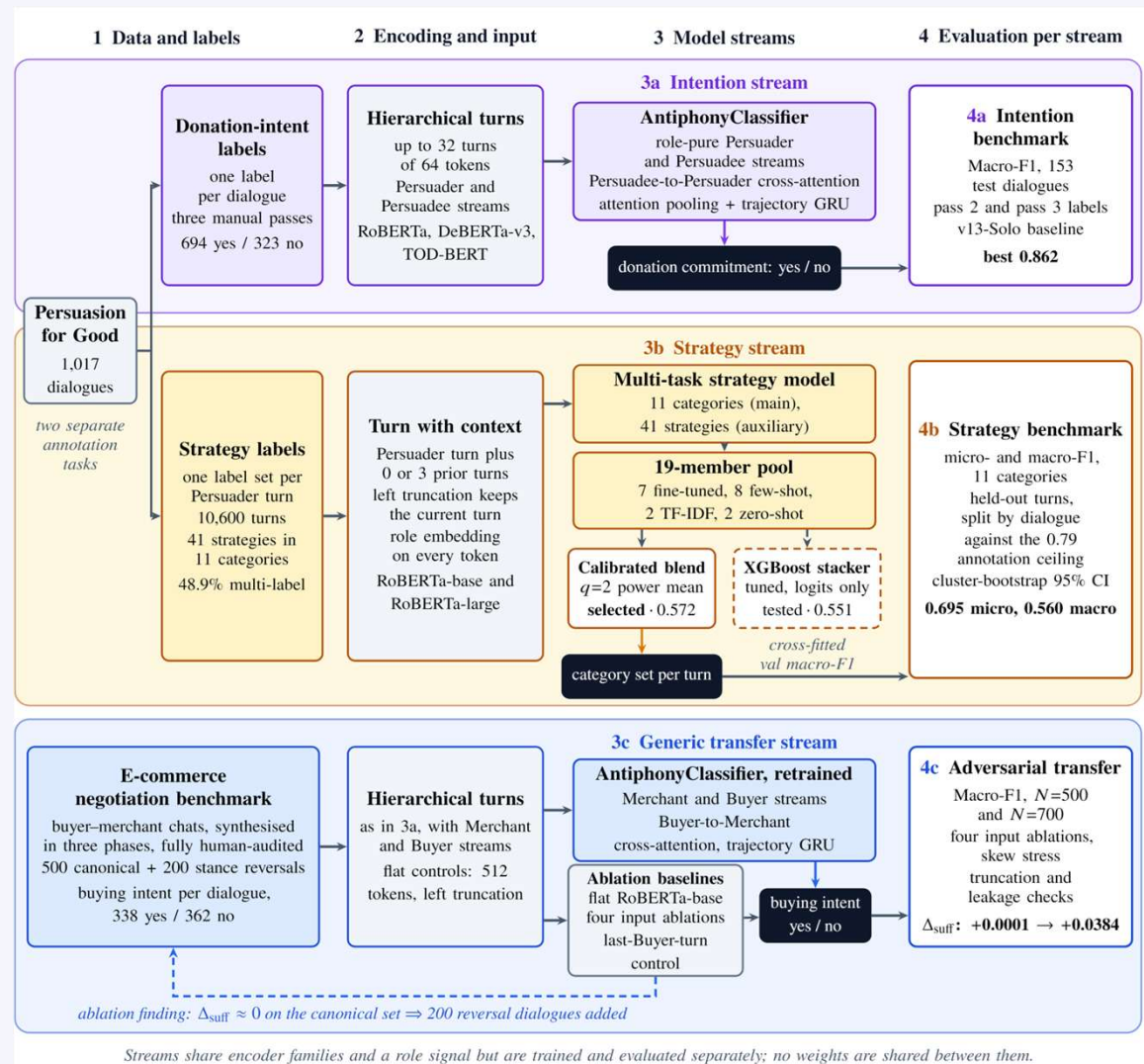
3b **Strategy** · P4G persuader turns

3c **Transfer** · e-commerce negotiation

COLUMNS · FOUR STAGES

1 Data and labels → 2 Encoding and input
→ 3 Model streams → 4 Evaluation per stream

Streams share encoder families and a role signal. No weights are shared between streams.



Rebuilding the Intention Ground Truth

Annotation pass

Pass 1 Manual, two fields

pilot of 200 dialogues, 50 per annotator, drift check then about 200 per member, pooled to 1,017 fields: `binary_label` and `modifier`
why: the corpus donation fields contradict the transcripts

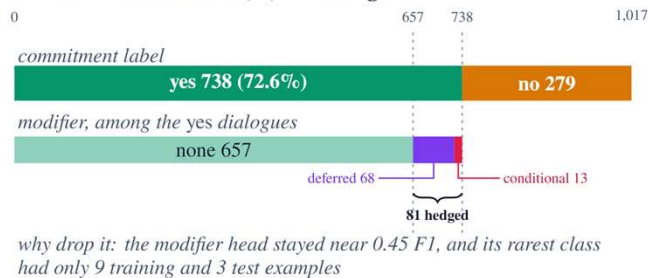
Pass 2 Plain binary

`binary_label` kept exactly as annotated
`modifier` dropped as a target, on the supervisor's direction

Pass 3 Strict commitment

yes only for a clear, unconditional commitment
any hesitation, condition or deferral counts as *no*
"I'll donate \$1 now, and I might give more later" is now *no*

Donation-intent labels, 1,017 dialogues



Used for

First-Pass Benchmark

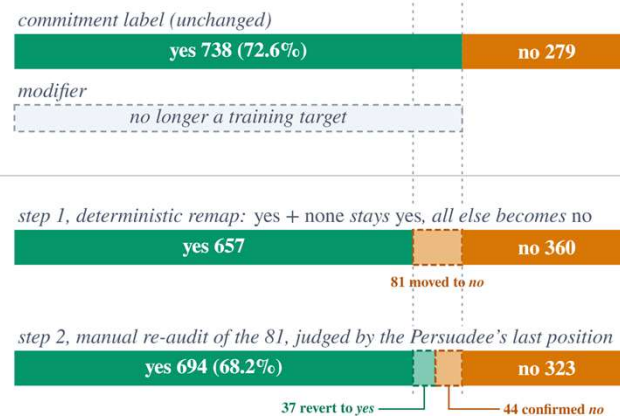
Dual-head models. Reliability check: a blind second labelling of 154 dialogues agreed on 142 (92.2%); the 12 disagreements were written back into the guideline.

Second-Pass Benchmark

Decoupled models. Without modifier gradients, the shared representation no longer trades binary accuracy against a data-starved head.

Third-Pass Strict Benchmark

Final labels. Used for the terminal v13-Solo and v14-Antiphony reruns.



PASS 1 · MANUAL, TWO FIELDS

738 yes / 279 no

Modifier: none 657, deferred 68, conditional 13

PASS 2 · MODIFIER DROPPED

Labels unchanged

On the supervisor's direction; the modifier never broke 0.45 F1

PASS 3 · STRICT

694 yes / 323 no

Any hedge or deferral counts as no

Rebuilding the Intention Ground Truth: Pass 3



92.21%

agreement with an independent second pass, 142 of 154 dialogues

THE 12 DISAGREEMENTS · FOUR BOUNDARY CASES

- i Ambiguous temporal deferrals: “I’ll donate through their website tonight”
- ii Polite deflection read as consent: “Sure, that sounds like a great cause”
- iii Conditional matches: “I’ll give \$1 if you give \$1”
- iv Unconfirmed donation queries

Strategy Taxonomy and Annotation Pipeline

11 41

categories strategies

IN CONTEXT

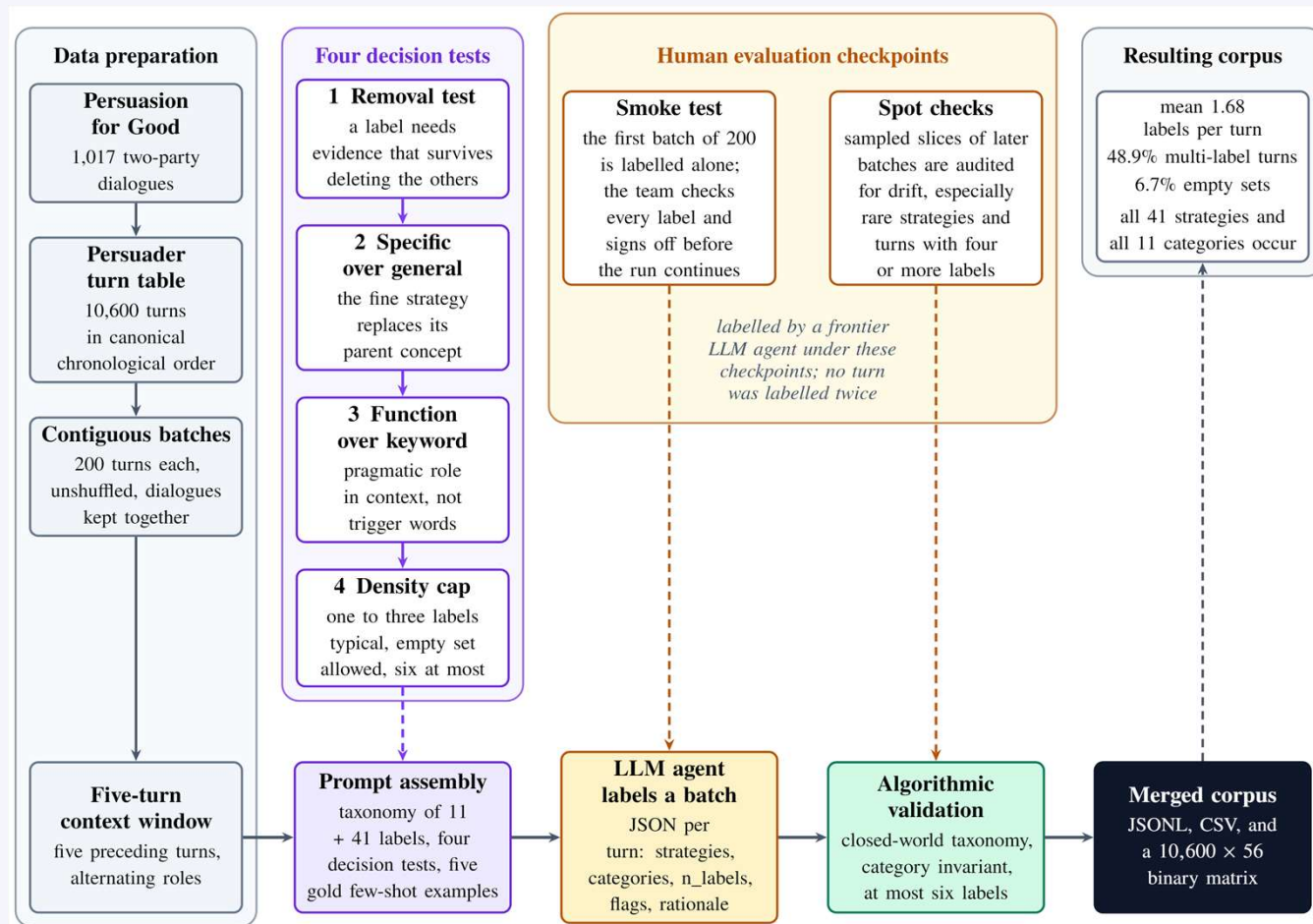
5 preceding turns, 200-turn batches

LLM AGENT

Claude 5 Sonnet labels each turn

HUMAN CHECKPOINTS

Smoke test and spot checks; pipeline.py validates every batch



Strategy Annotation: Four Decision Tests

01

Removal test

Keep a label only if its evidence survives deleting the others

02

Specific over general

The fine strategy replaces its parent concept

03

Function over keyword

Label the pragmatic role in context, not trigger words

04

Density cap

One to three labels typical, empty set allowed, six at most

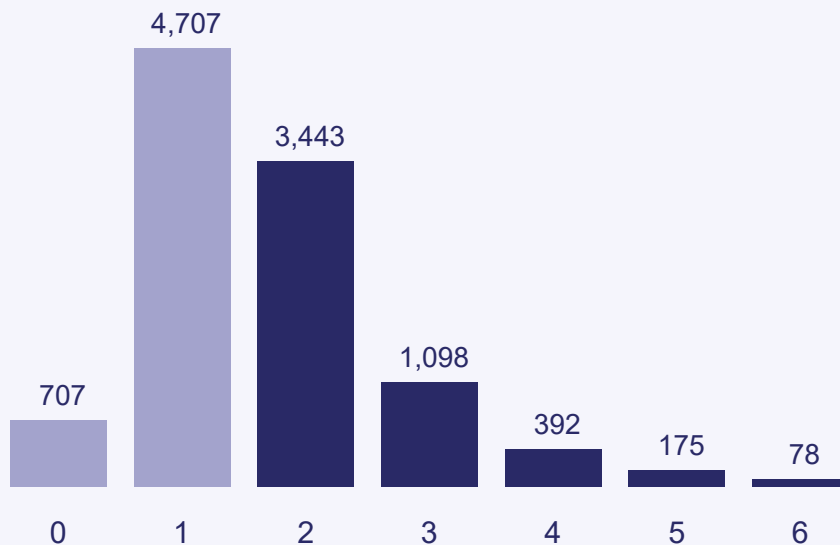
10,600 × 56 output matrix, zero schema violations

Strategy Corpus Statistics

48.9%

of turns are multi-label · mean 1.68,
6.7% empty

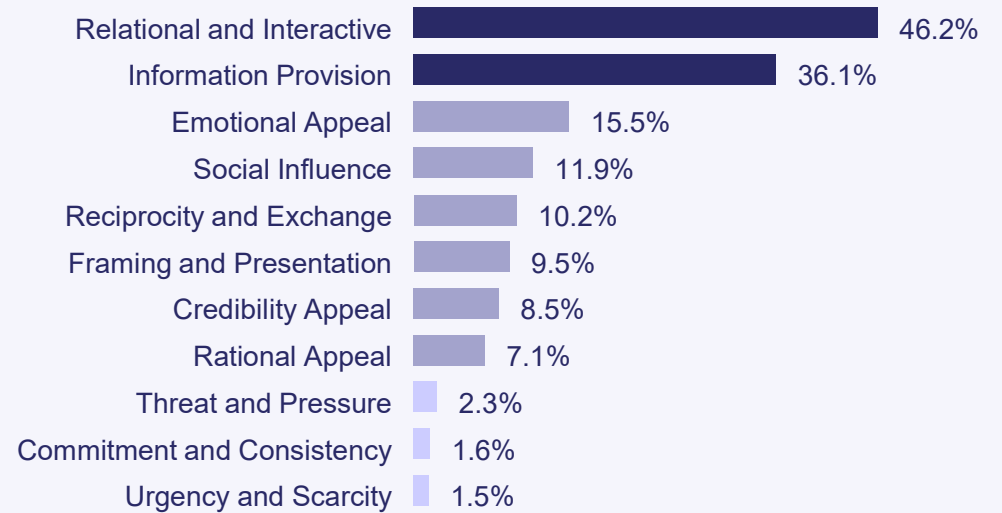
Turns by number of strategy labels (2 or more in dark)



46.2%

Relational and 36.1% Information dominate

Share of turns by category · Threat, Commitment, Urgency each under 3%

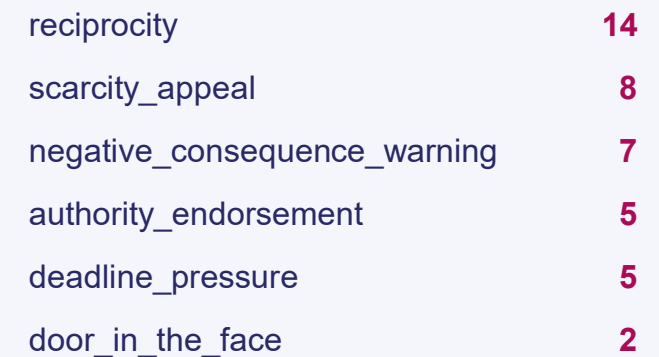


Strategy Corpus Statistics: The 41-Strategy Long Tail

Most frequent strategies · turns out of 10,600



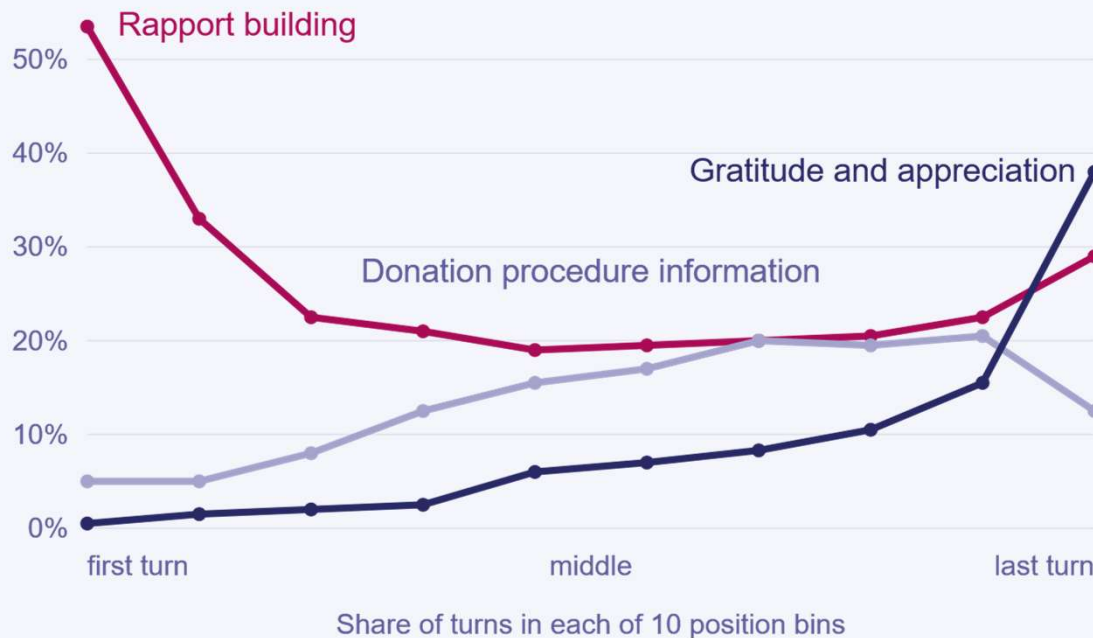
Rarest strategies · turns



rapport_building alone labels 27.0% of turns. Six strategies appear in 14 turns or fewer.

Strategy Corpus Statistics: Timing and Ceiling

Strategies follow the dialogue's shape: rapport comes early, procedure and gratitude come late.



HUMAN AGREEMENT CEILING

Same annotator **0.871**

Cross-annotator **0.657**

Consensus ceiling **≈ 0.79**

Micro-F1 above ≈ 0.79 would mean fitting one annotator's habits.

Rater effect: one segment averaged 2.38 labels per turn, another 1.24.

Synthetic E-commerce Negotiation Benchmark

PHASE 1 · PERSONA SYNTHESIS

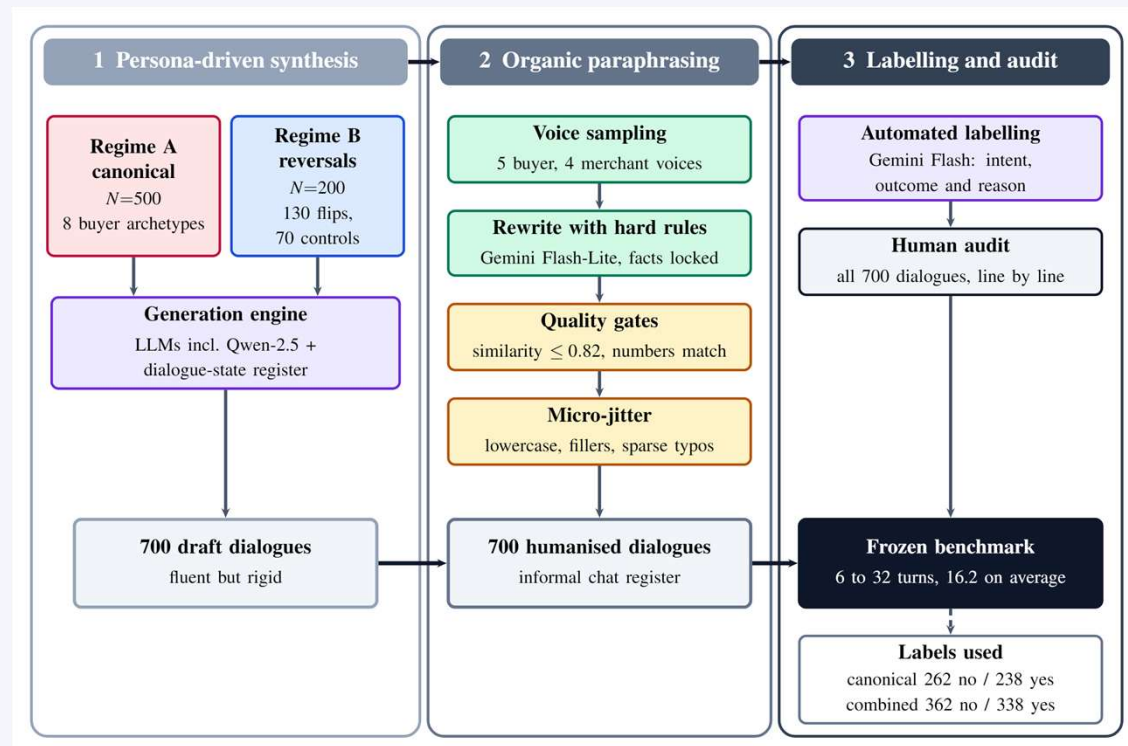
Regime A canonical, N = 500, 8 buyer archetypes. Regime B, N = 200: 65 + 65 reversals, 70 controls.

PHASE 2 · PARAPHRASING

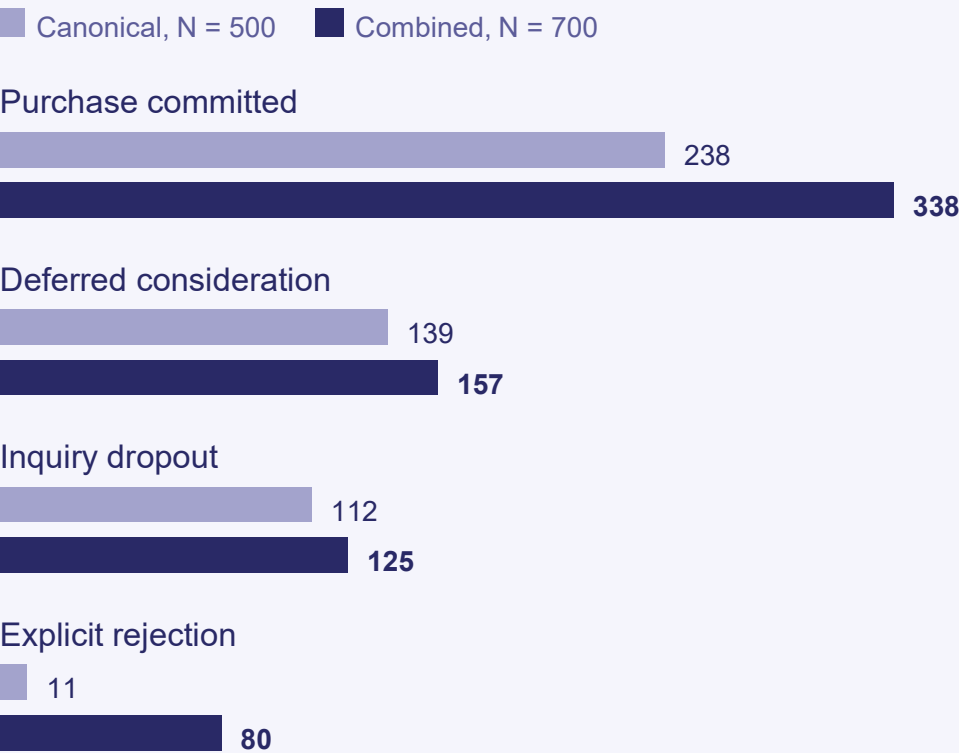
Gemini, facts locked, similarity gate 0.82, micro-jitter

PHASE 3 · LABELS AND AUDIT

Automated labelling plus a 100% human audit



Negotiation Benchmark: Counts and Rationale



CANONICAL
262 / 238
no / yes

COMBINED
362 / 338
no / yes

REVERSALS
100 / 100
no / yes

TURNS
16.2
mean, range 6 to 32

WHY SYNTHETIC

Real merchant chats are private. Canonical data was “solvable from the last line”, so 200 reversals were added.

— PART 3 OF 5

Models

- 3.1 Architectural evolution
- 3.2 Antiphony: the idea
- 3.3 Antiphony: inside
- 3.4 Strategy classifier ensemble



How We Got to Antiphony

PEAK TEST MACRO-F1 PER STAGE

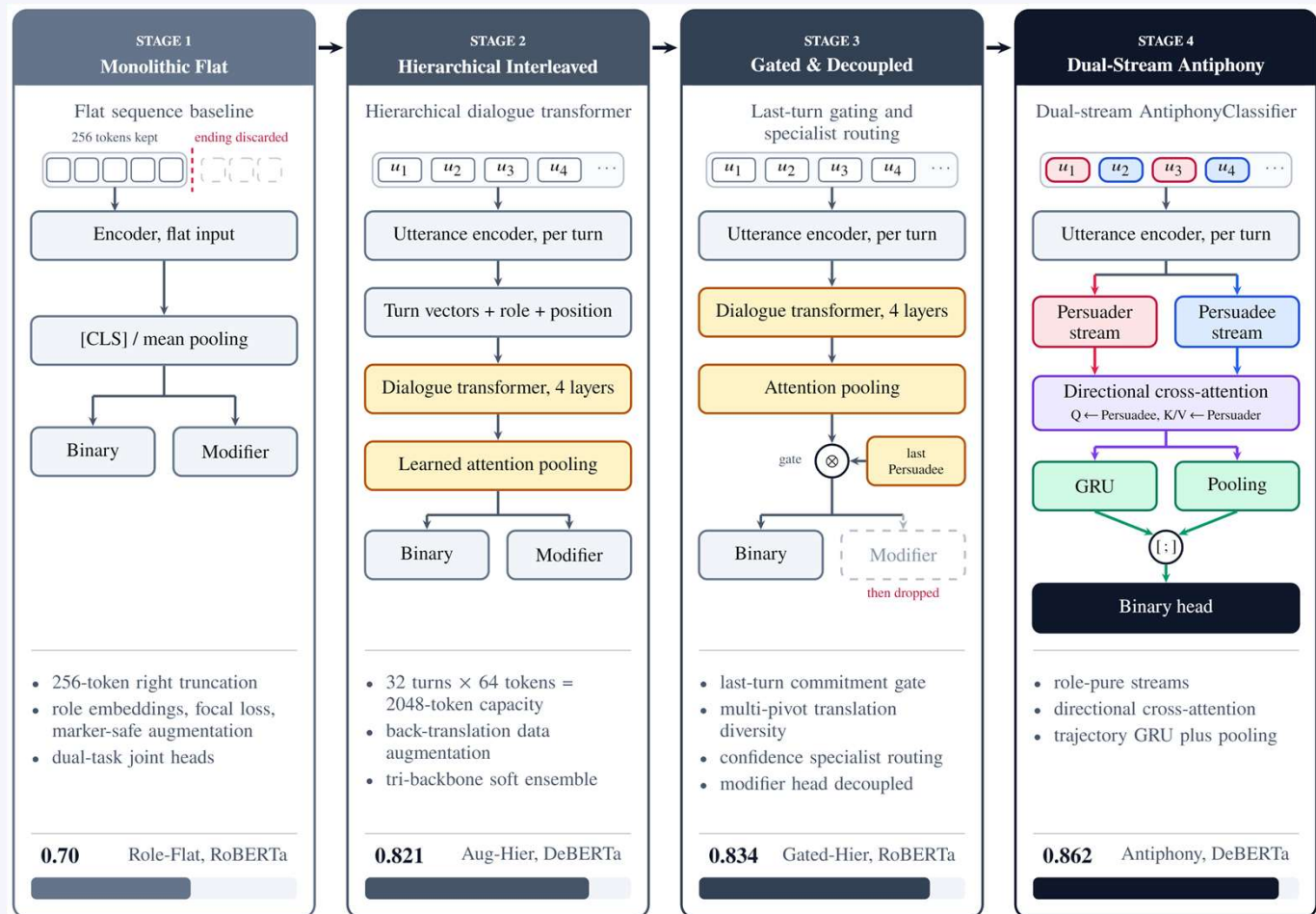
1 · Flat **0.70**

2 · Hierarchical **0.821**

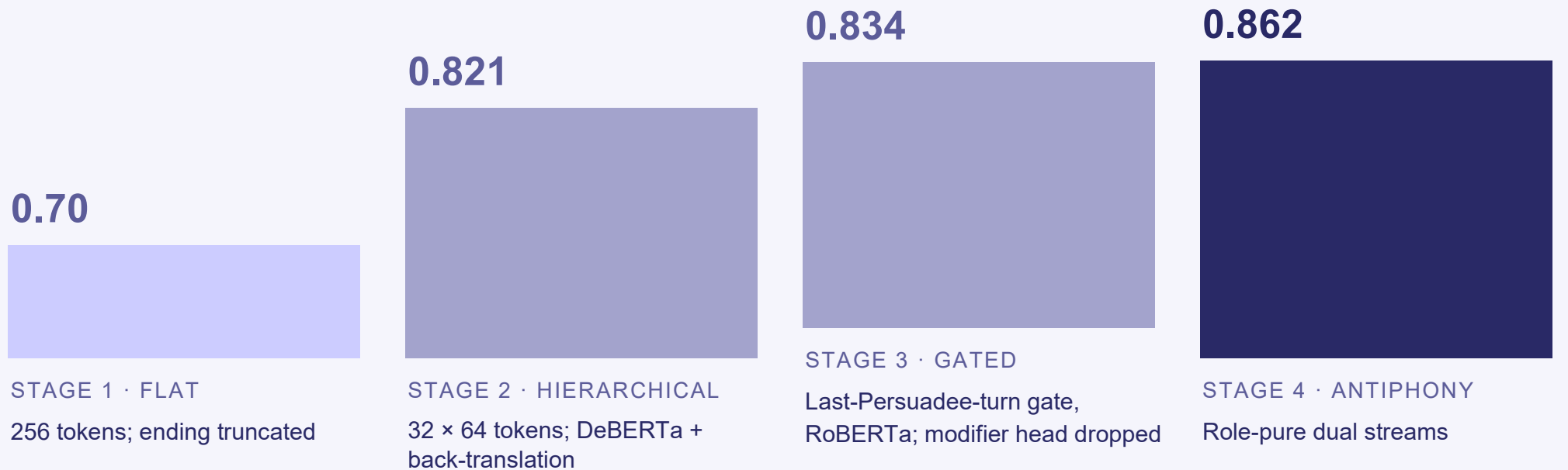
3 · Gated and decoupled **0.834**

4 · Antiphony **0.862**

“Each stage fixed the previous stage’s failure. The remaining failure was that the two speakers were mixed together.”



How We Got to Antiphony: Stage by Stage



STAGE 1 · THREE BUGS FOUND AND FIXED

Checkpoint metric

Chosen on pos_label="yes" F1, so an all-yes model scored 1.0

Gradient spikes

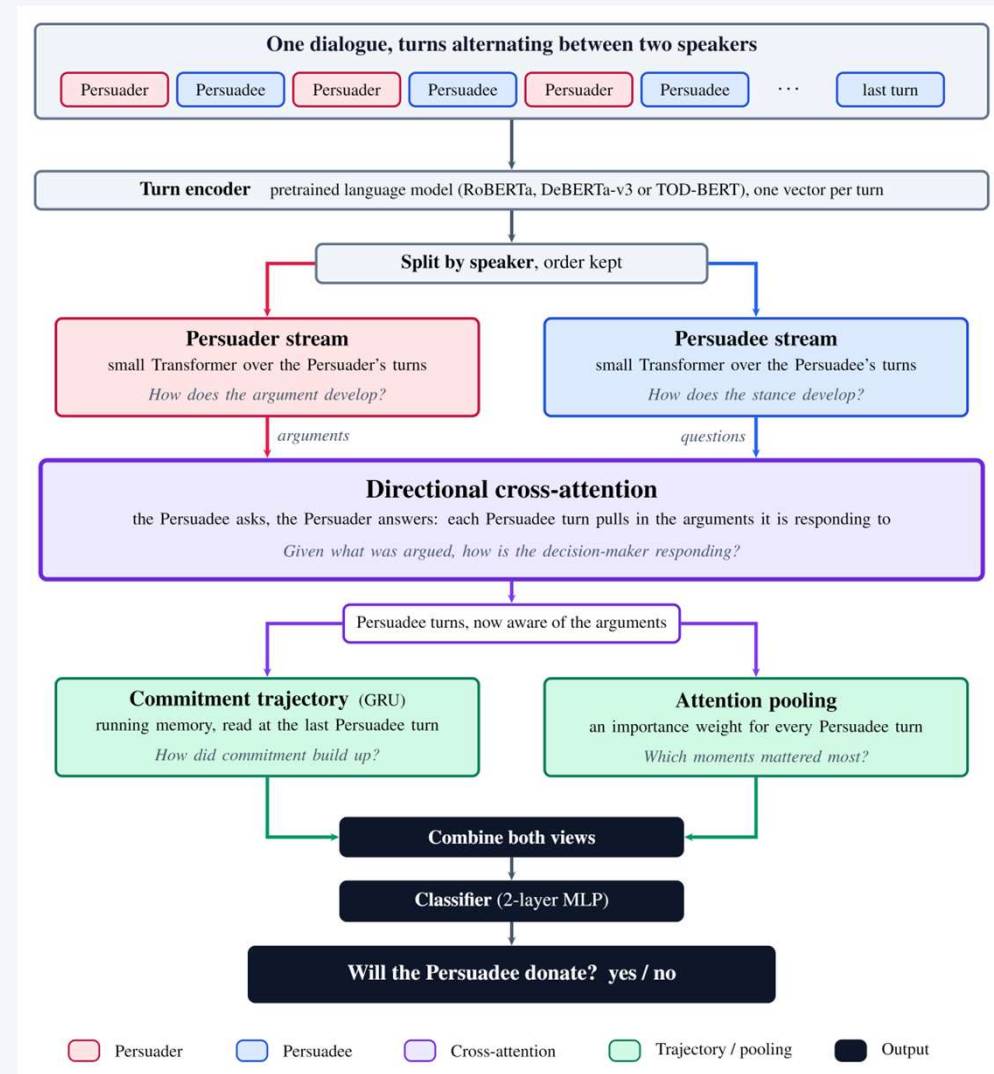
Raw inverse-frequency weights on 13 conditional examples

Glued speaker tags

57.5% glued to newlines, corrupting 15.9% of augmented rows

Antiphony: The Idea

- 1 Encode each turn
- 2 Split the turns by speaker, order kept
- 3 Each speaker gets its own small Transformer
- 4 Directional cross-attention: the Persuadee asks, the Persuader answers
- 5 A GRU tracks commitment in turn order; attention pooling picks the salient turns
- 6 Both views are combined; a classifier predicts yes / no



Antiphony: Directional Cross-Attention

“Given what was argued, how is the decider responding?”

Persuadee → queries

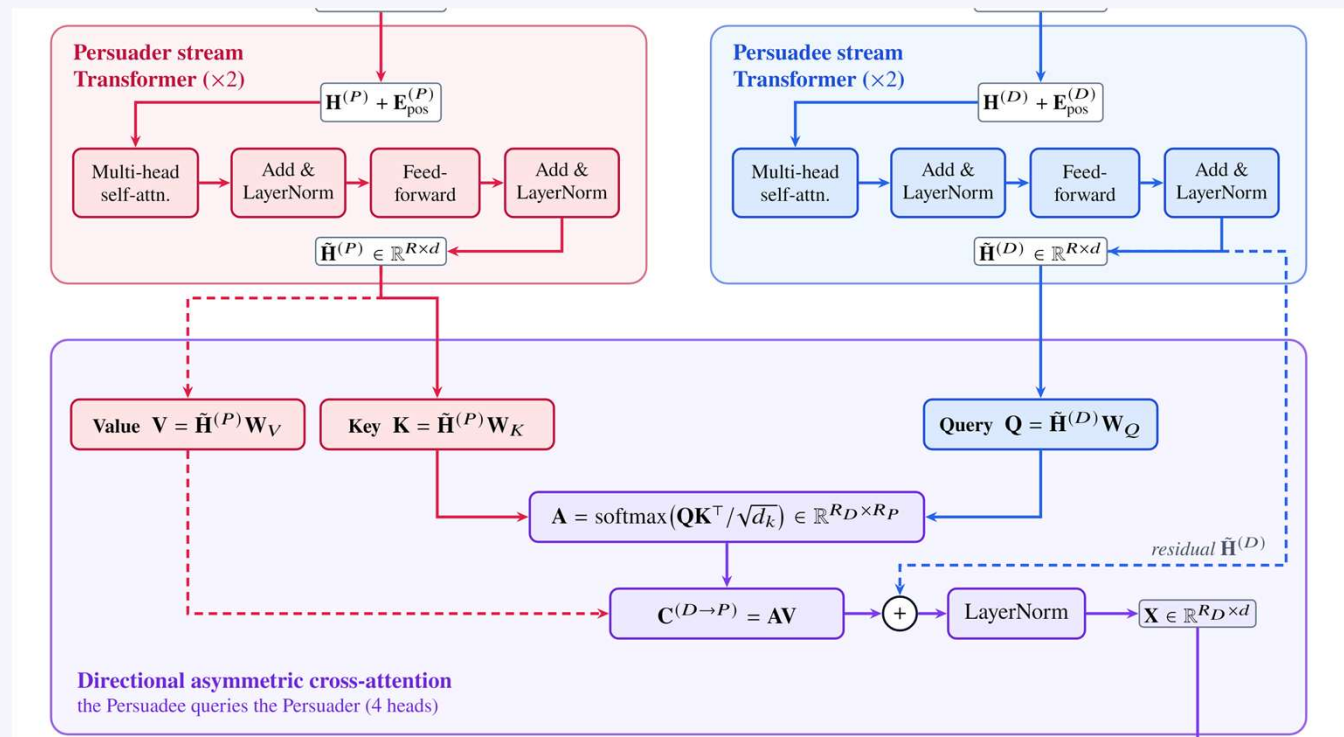
$Q = \tilde{H}^{(D)} W_Q$. The decider asks.

Persuader → keys, values

K, V from $\tilde{H}^{(P)}$. The arguments answer.

Output

$C = AV$, plus a residual from the Persuadee stream, 4 heads



Antiphony: Inside

A $Q = \tilde{H}^{(D)} W_Q$, with K, V from $\tilde{H}^{(P)}$

A $A = \text{softmax}(QK^T / \sqrt{d_k})$

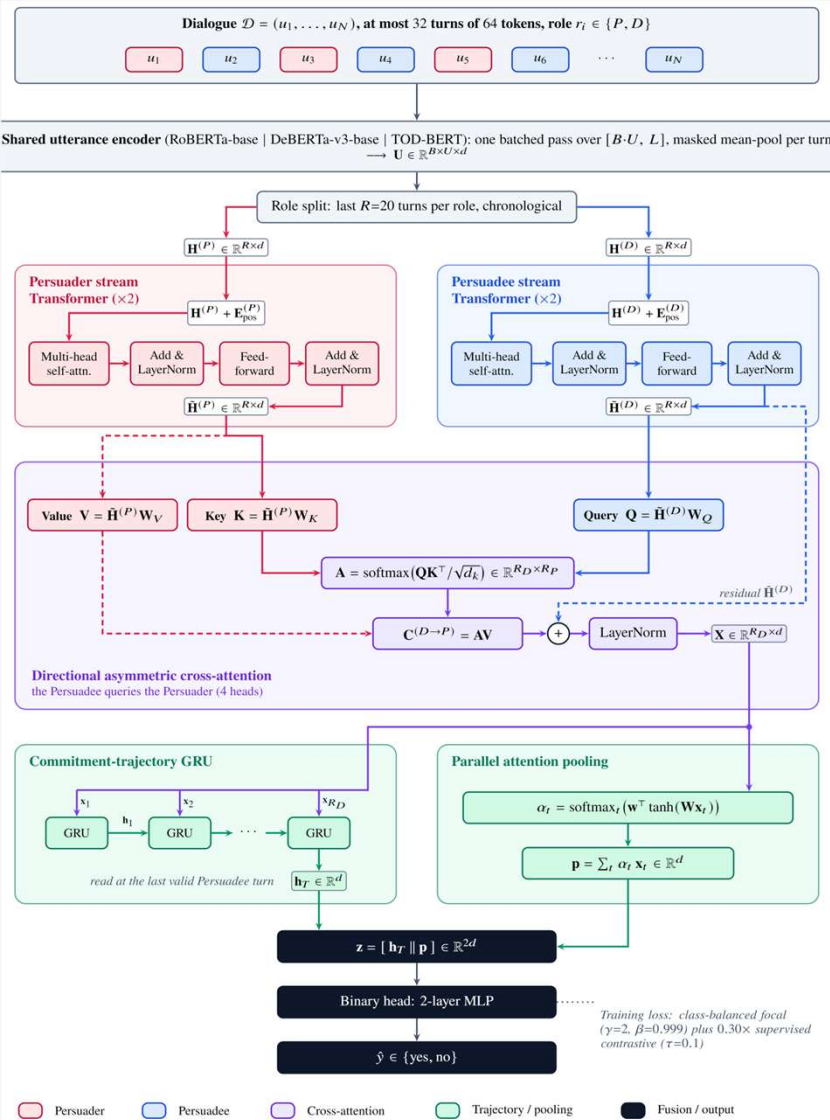
B $z = [h_T; p] \rightarrow 2\text{-layer MLP}$

LOSS

CB-Focal ($\gamma = 2, \beta = 0.999$) + $0.3 \times \text{SupCon} (\tau = 0.1)$

STREAMS

Two Transformer layers per role, last R = 20 turns per role, chronological



Antiphony: Design Refinement

SEQUENCE BUDGET

2,048

tokens · $U = 32 \text{ turns} \times L = 64$

covers 99.7% of tokens

MEMORY · T4 16 GB

B = 2

plus gradient checkpointing

about 60% less memory

ROLE WINDOW

R = 20

turns per role

covers 98.8% of dialogues

INFERENCE

45 ms

per dialogue on GPU

380 ms on CPU

Strategy Classifier: The 19-Member Ensemble

PIVOT

0.368

macro-F1 on all 41 strategies

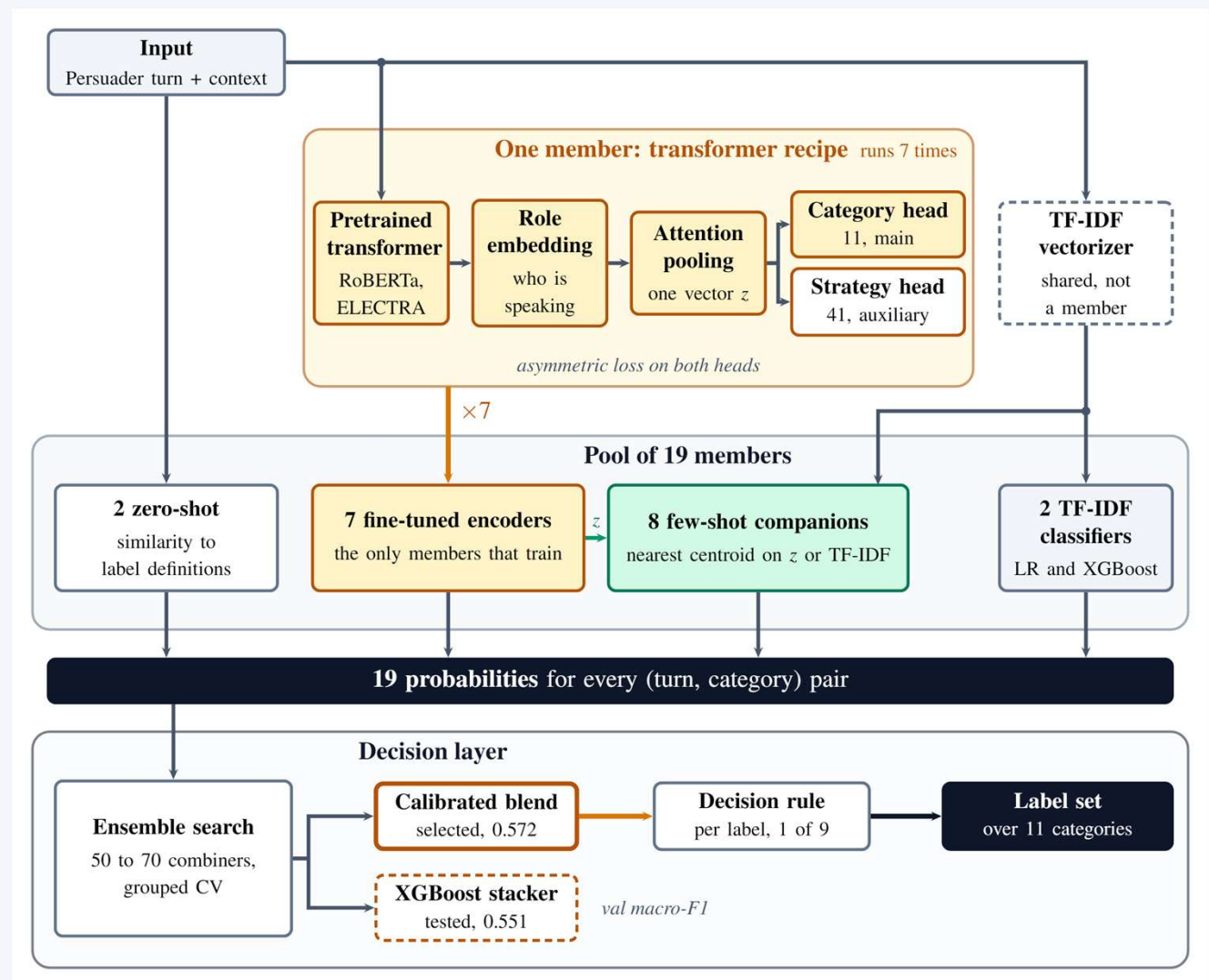
So: 11 categories as the main target, 41 strategies as a helper

POOL OF 19

7 fine-tuned encoders · 8 few-shot companions
2 TF-IDF · 2 zero-shot

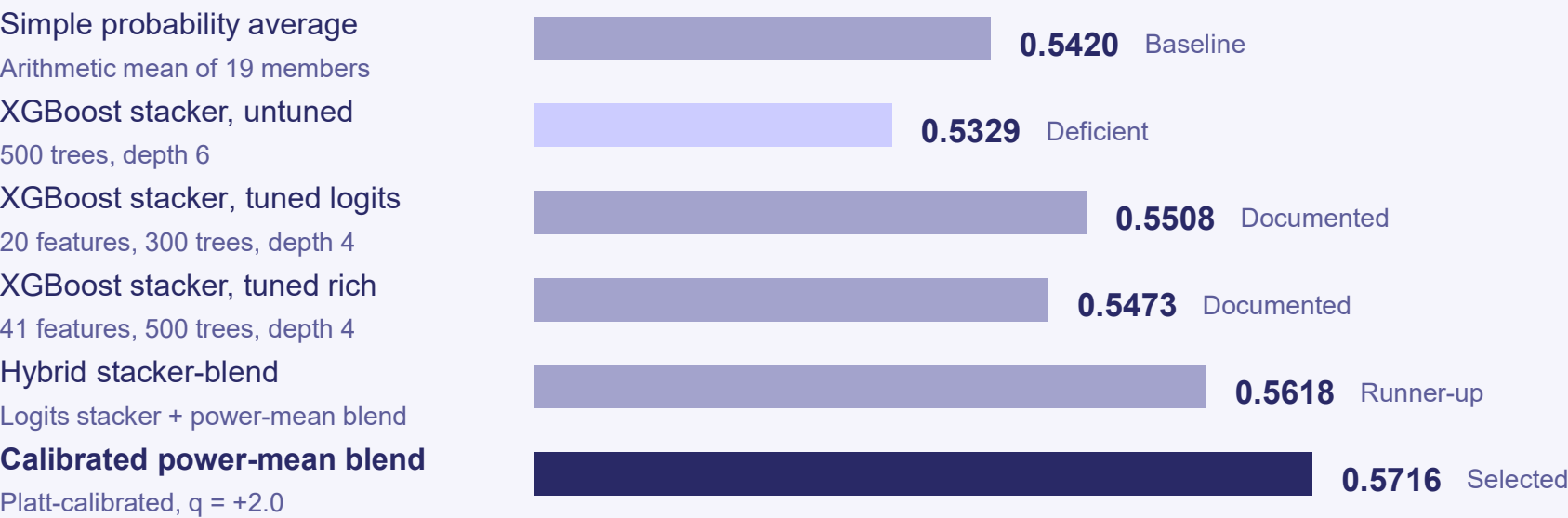
TRAINING

Asymmetric Loss on both heads



Strategy Classifier: Decision Layer

Validation macro-F1 · Table 5.9 · bars start at 0.50



50–70

candidate combiners
searched with
grouped cross-
validation

— PART 4 OF 5

Results

4.1 Intention

4.2 Strategy

4.3 Transfer

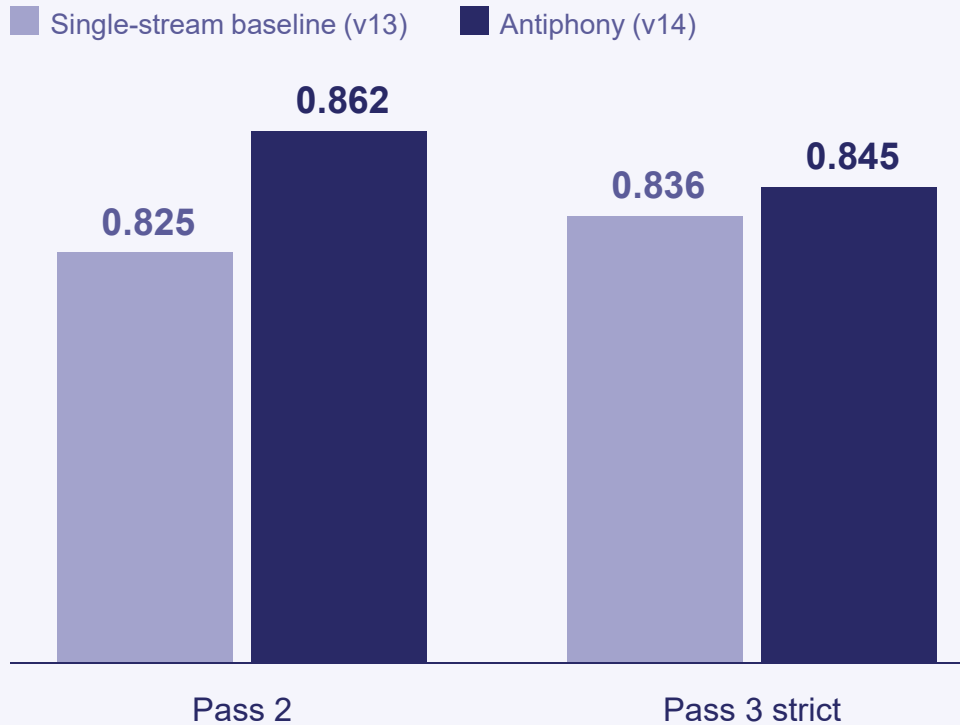
4.4 Sufficiency gap

4.5 Robustness and forensics

4.6 Havn demo



Intention Results



Test Macro-F1 · axis starts at 0.70

PASS 3, FINAL MODEL (ROBERTA)

0.78 0.91 86.9%

no F1

yes F1

accuracy

Test n = 153 dialogues · 70 / 15 / 15 split at dialogue level · Pass 2
Antiphony 0.860 with tuned threshold · baseline 0.829 tuned on Pass 3

“Antiphony wins on both label sets, even against the baseline’s best threshold.”

Strategy Results Against the Human Ceiling

THE ARC · TABLE 5.7

41 strategies

0.368

11 categories

0.556

v18 blend

0.695 / 0.560

macro-F1; last row micro / macro

MICRO-F1 AGAINST THE HUMAN CEILING

89%

0.706 micro-F1 = 89% of what humans agree on

0.706 best (v17)



0

0.79 human ceiling

Blend 0.572 vs XGBoost 0.551 are validation macro-F1, not test. v17 (0.706 / 0.564) and v18 (0.695 / 0.560) are statistically tied: bootstrap 95% intervals overlap almost completely.

Transfer: Do Models Need the Whole Dialogue?

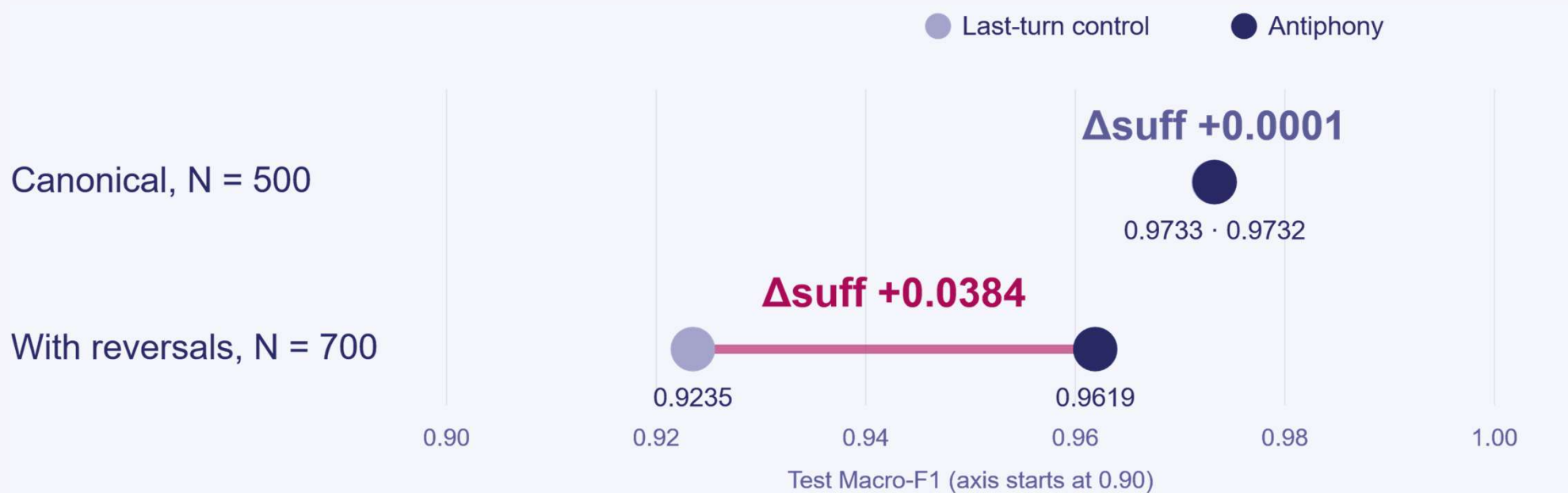
FLAT ROBERTA · MACRO-F1	LTR	LTR+M	FHO	PRO
Canonical · N = 500 · test n = 75	0.933	0.933	0.771	n/a
Augmented · N = 700 · test n = 105	0.952	0.914	0.848	0.950
Reversal subset · test n = 30	1.000	0.967	0.866	0.958

LTR last turn removed **LTR+M** plus trailing merchant confirmations **FHO** first half only **PRO** pre-reversal only

Removing the last turn barely hurts. Even the pre-reversal input scores 0.950: merchant concessions give the outcome away early.

0 / 20 stance-flip errors

The Sufficiency Gap



“When the ending stops giving the answer away, the last-turn model loses 0.05 and Antiphony holds. The gap is the trajectory.”

Robustness and Forensics: Skew Stress

0.933–0.973 **47.7% → 9.8%**

Macro-F1 range across the stress arms

positive training examples



Robustness and Forensics: Tokenizer Truncation

30.4%

of dialogues (152 of 500) lost their endings
to default right-truncation

0.813 → 0.933

after switching to left-truncation (+0.12)

*“Our biggest ‘gap’ at one point was a
tokenizer default, not the model.”*

Test Macro-F1, canonical benchmark

Full model (DeBERTa-v3, dual-stream)



Last turn removed · left-truncation



Last turn removed · right-truncation (default)



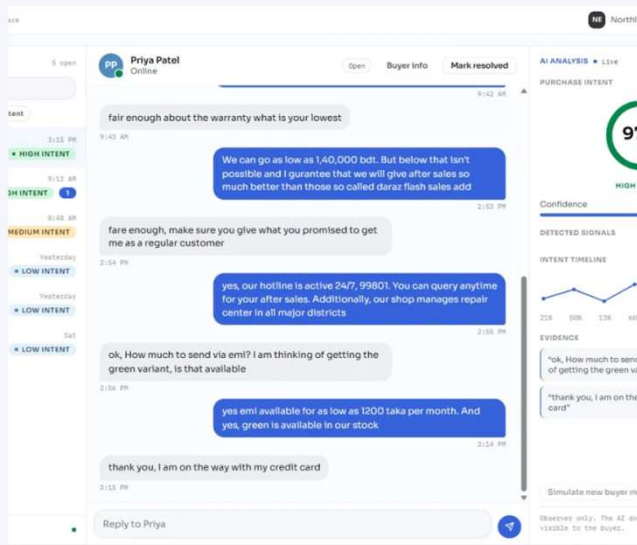
First half only



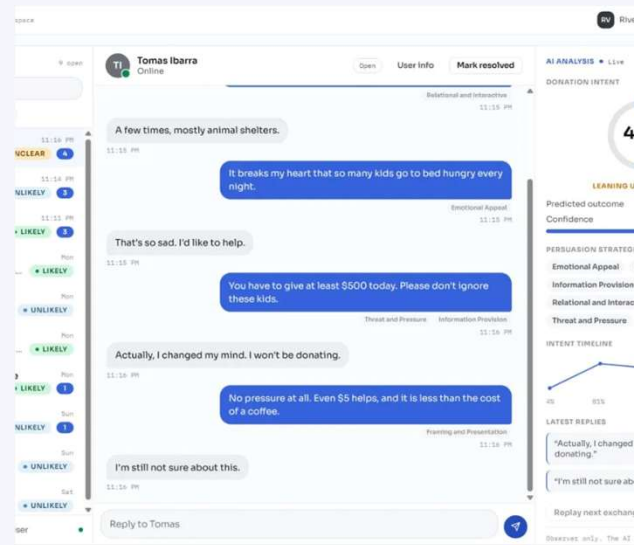
Apparent necessity gap: 0.1600 → 0.0402

Havn Demo

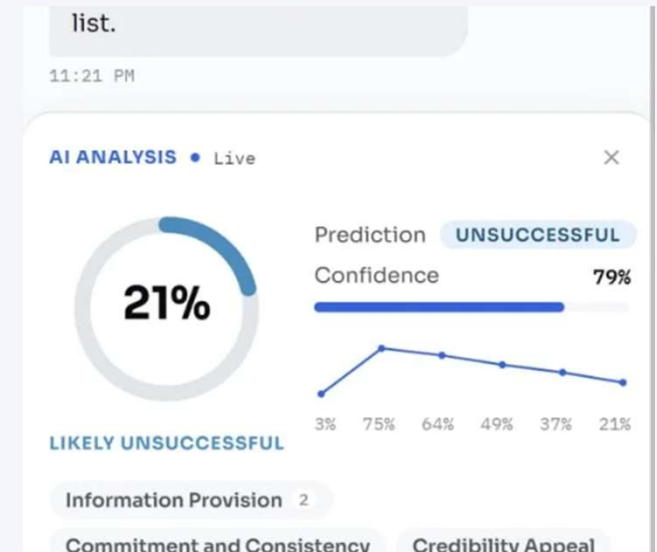
“Observer only. It reads the chat, scores intent turn by turn, and tags strategies. It never replies.”



97% Buyer-seller intent, rising per turn



49% Donation intent falls after a stance reversal



Mobile view

— PART 5 OF 5

Close

5.1 What we learned

5.2 Conclusion and future work

5.3 Questions



What We Learned

01	Modifier head dropped	because of data sparsity 9 training examples for conditional; it never broke 0.45 F1
02	Categories over strategies	because of the long tail 0.368 macro-F1 on 41 strategies; categories reached 0.556
03	XGBoost stacker rejected	in favour of the calibrated blend 0.5508 against 0.5716 validation macro-F1
04	Right-truncation replaced	by left-truncation 30.4% of endings lost; necessity gap 0.1600 → 0.0402

“Each negative result is documented, not hidden.”

Conclusion and Future Work

RECAP

0.862 / 0.845

intention Macro-F1

0.706

strategy micro-F1, 89% of ceiling

+0.0384

sufficiency gap on reversals

LIMITATIONS

Only 153 test dialogues

A single seed and split

English only

The negotiation benchmark is synthetic

FUTURE WORK

Bengali / Banglish with BanglaBERT

Early-warning signal from the GRU state

Multi-seed significance tests

Back-translation on the persuader side only



ANTIPHONY · CSE 450 · GROUP 10

Thank you

Questions and discussion

0.862

intention Macro-F1

0.706

strategy micro-F1

+0.0384

sufficiency gap

SCAN FOR THE LIVE DEMOS



Persuasion

Donation intent & strategy



Buyer-Seller

Negotiation intent